

Technology Transformation

Read More



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Objective

Digitalization, automating tasks & improving efficiency

Projects

- **Worker Timesheet System** – Automated tracking of worker presence
- **Web Scraper for Tenders** – Extracting and filtering tender listings
- **Tender Tracking System** – Organizing deadlines with Google Sheets & Calendar



Pain Point:

- Workers manually filling timesheets -> prone to errors
- Inaccurate records lead to overtime disputes and extra costs

Proposal 1 - Worker Timesheet System

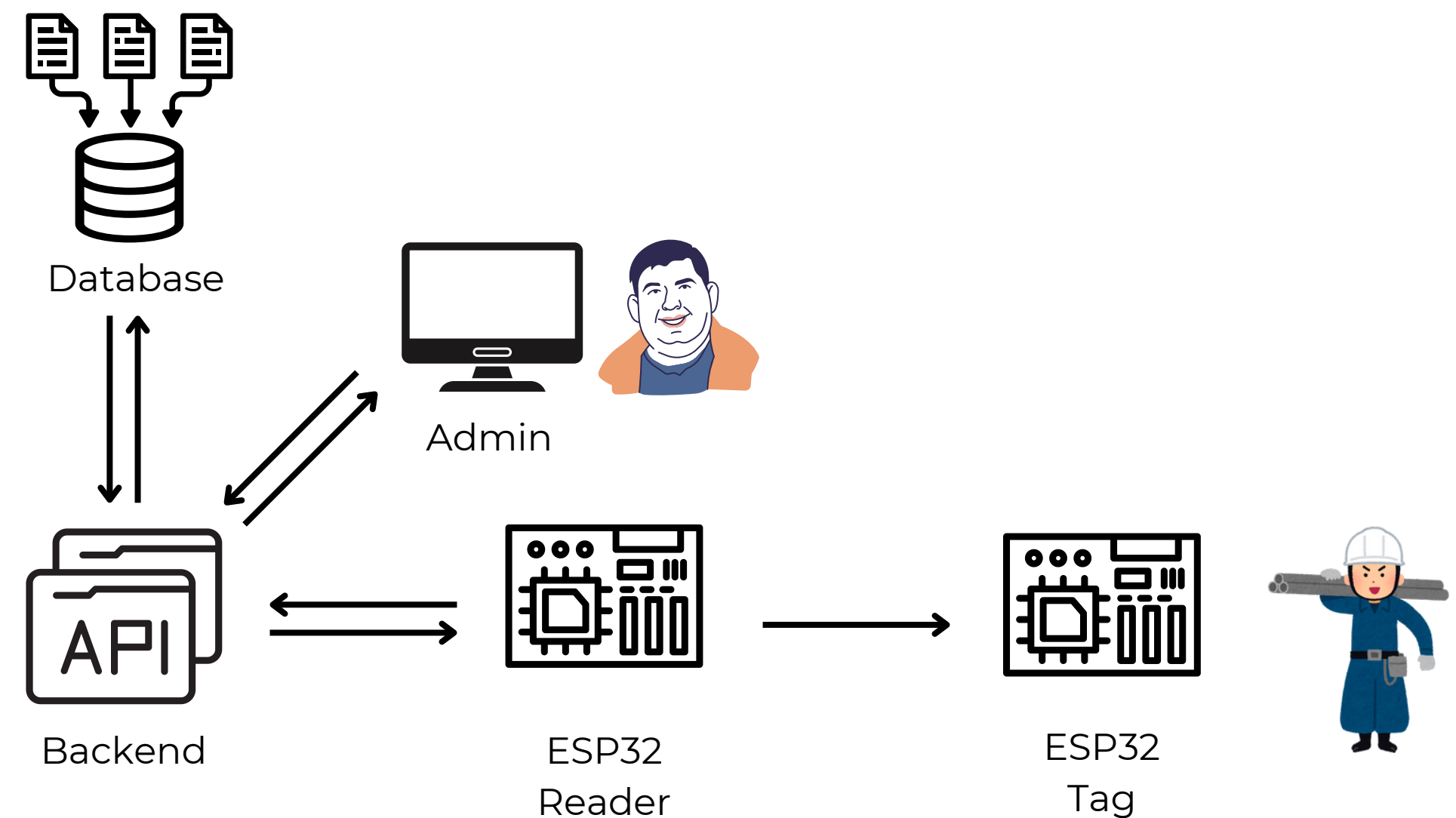
Proposal 1 - Worker Timesheet System

Instead of relying on workers to fill in timesheets, propose using a small device called ESP32 that will automatically detect when workers arrive and leave the site.

The ESP32 Reader will be placed at the worksite. When a worker arrives, the device detects their beacon/ESP32 tag's Bluetooth and records the check-in time.

When they leave, it records the check-out time.

All this information is sent to a central system where admins can see accurate timesheets.



Tech Stack

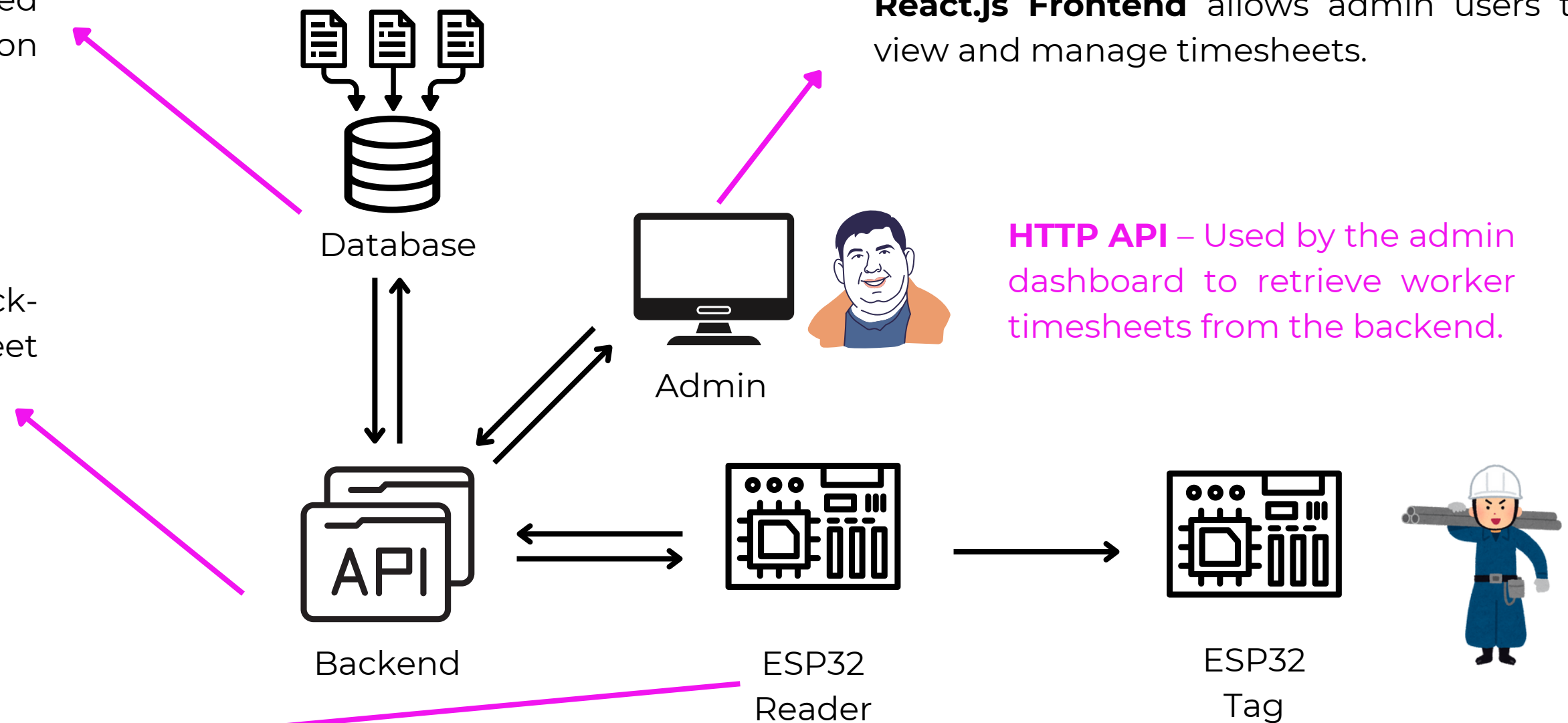
MySQL Database for storing data related to users, hardware on site and connection times

Node.js Runtime handles worker check-in/check-out data and timesheet processing

ESP32 detects worker phones via Bluetooth and logs attendance to MySQL Database

React.js Frontend allows admin users to view and manage timesheets.

HTTP API – Used by the admin dashboard to retrieve worker timesheets from the backend.



Pain Point:

- Manual Tender Checking is Time-Consuming -> Staff had to visit websites daily to check for new tenders.

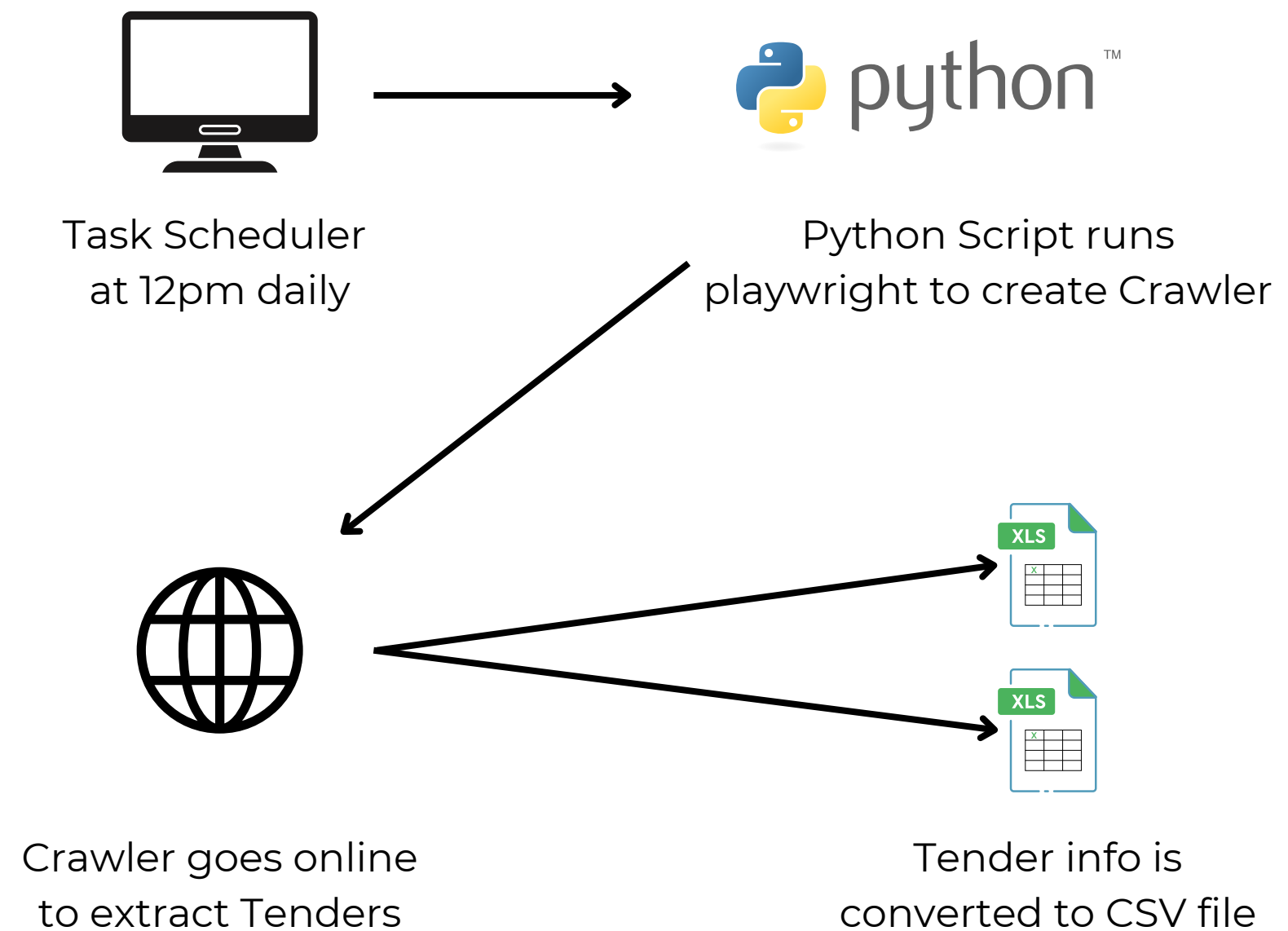
Proposal 2 - Web Scraper for Tenders

Proposal 2 - Web Scraper for Tenders

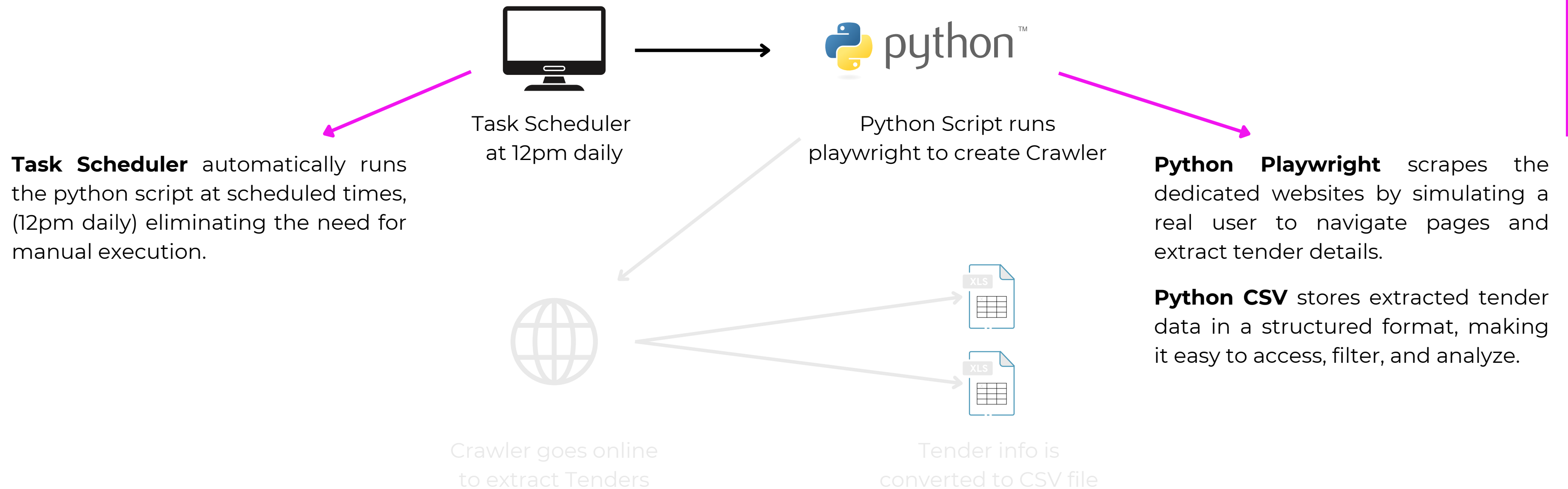
To solve the issue of manual tender checking, a web scraper utilizing python library, Playwright was developed. The scraper automates the process of visiting EM Services and GeBIZ site (<https://www.emservices.com.sg/tenders/>) and (<https://www.gebiz.gov.sg/ptn/opportunity/BOListing.xhtml?origin=menu>), extracting tender details, and filtering them for easy access.

Instead of relying on staff to check tenders daily, the scraper runs on a scheduled basis and collects new tenders automatically. This reduces manual effort, minimizes human errors, and ensures no tender opportunities are missed.

By using Task Scheduler instead of a dedicated server, we achieved the same automation while keeping costs low. This makes the solution efficient, scalable, and cost-effective.



Tech Stack



Pain Point:

- Physical calendar made it difficult to keep up with multiple tender deadlines
- No centralized system to track tender submission statuses.

Proposal 3 - Tender Tracking System

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To enhance efficiency and maximize reusability, a tender tracking system was implemented using **Google Sheets** and **Google Calendar**.

Tender details are logged in Google Sheets, while deadlines are automatically added to Google Calendar for reminders.

This system ensures that tenders are well-organized, easily accessible, and effectively tracked, reducing the risk of missed deadlines.



Google Sheets



Google Apps Script



Google Calendars

Tech Stack



Google Sheets



Google Apps Script



Google Calendars



Google Sheets records and tracks tender details in a structured format.



Google Apps Script automates the process of syncing deadlines from Google Sheets to Calendar.



Google Calendar displays tenders in a calendar format, retaining the calendar view while also providing automatic reminders.